

INFORMED CONSENT FORM

Audio and Video Recording of Research Participation

Study Title: Expert Validation of Pedagogical Recommendation Cards Derived from Discourse-Level Pedagogical State Tracking

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1. Purpose of the Study

You are being invited to take part in a research study examining whether recommendations generated from a computational model of classroom discourse match the independent professional judgement of experienced teachers. The purpose of this session is to understand how educators evaluate short coaching suggestions written for real classroom lesson excerpts, and whether such suggestions are clear and useful in practice.

Your participation will help evaluate a research tool intended to support teacher coaching and professional development. Your specific responses will not be used to evaluate your own teaching.

2. Audio and Video Recording

With your permission, this session will be audio and/or video recorded. Recording allows the research team to accurately capture your spoken responses and reasoning for later analysis. Recordings will be used only for the following purposes:

- Transcription of your verbal responses for research analysis.
- Review by members of the research team to ensure responses are coded accurately.
- Short, anonymized audio or video clips may be used in academic presentations or publications only with your separate, specific permission (see Section 6).

Recordings will not be posted publicly, shared with your employer or school administration, or used for any purpose other than those listed above, without your further written consent.

3. Voluntary Participation and Right to Withdraw

Your participation in this study is entirely voluntary. You may decline to answer any question, skip any part of the session, or stop participating at any time, without giving a reason and without any penalty or effect on your professional standing.

You may also choose to participate in the study without being audio or video recorded. If you prefer this option, the researcher will instead take written notes during the session. Please indicate your preference in Section 7 below.

If you withdraw during the session, any recording made up to that point will be deleted at your request, unless you indicate that partial data may still be used for research purposes.

4. Confidentiality and Data Storage

- Your name will not be used in any research report, publication, or presentation. You will be identified only by a participant code (e.g., "Teacher 3").
- Audio/video recordings and transcripts will be stored on a password-protected device accessible only to the research team.

- Recordings will be retained for [X years / until study completion plus X years, per your institution's data retention policy] and then securely deleted.
- Any classroom transcript excerpts shown to you during the session have already been de-identified (no student, teacher, or school names).

5. Additional, Separate Permissions

The permissions below are optional and independent of your decision to participate in the study. You may participate fully while declining any or all of the items below.

- ☒ I agree that anonymized audio clips of my voice may be used in academic conference presentations.
- ☐ I agree that anonymized video clips of me may be used in academic conference presentations.
- ☐ I agree that de-identified quotes from my responses may be reproduced in published research articles.
- ☐ I agree that my anonymized data may be retained and reused in future related research studies.

6. Your Recording Preference

Please check one:

- ☐ I consent to being audio recorded during this session.
- ☐ I consent to being video recorded during this session.
- ☒ I consent to audio recording only (no video).
- ☐ I do NOT wish to be recorded; I agree to the researcher taking written notes instead.

Participant Consent and Signature

By signing below, I confirm that I have read and understood the information above, that my questions have been answered, and that I voluntarily agree to participate in this study under the recording preference I have indicated in Section 6.

Participant Name: _____

Gender: Male

Age: 43

Designation: PET

Institution(s): K V DRDO

Teaching subject: PHYSICS

Teaching experience (in years): 16yr

Participant Signature: _____

Date: _____

22/09/26

Researcher Name: _____

Researcher Signature: _____

Date: _____

Contact Email / Phone: viveku@iisc.ac.in/7905208259

Clip A

1. In the clip shown, is there anything the teacher is doing well?

- * Asking questions
- * Involving students' responses

2. In the clip shown, is there anything that the teacher is not doing well?

- * No well structured questions
- * Changed the framing of questions after receiving undesired answers
- * No visual / demonstration / activity to show the effect of indicators.

3. Suggest, if there is any scope of improvement

- * Visuals / PPT / Activity / Demonstration should be included
- * Questions should be well framed based on above activities / demonstration
- * Questions should be in inductive - deductive format so that students should achieve the generalisation at their own

Clip A

For each question, you will be shown multiple coaching notes written about the same clip. Please read all the options carefully and select **only one** option that, in your judgment, best fits what is actually happening in the specific transcript. There is no requirement that any option is correct. Choose the option that you think best matches the transcript based on your own judgment and expertise. If none of the options can be distinguished as a better fit, select “**Neither / Can’t tell.**”

Please select only one option for each question.

1a). The teacher should ask the students to recall and name the examples of indicators multiple times; this enables better retention due to constant repetition. ☐

1b). The teacher should avoid asking the students to recall the name of the synthetic indicator which promotes retention over understanding of the concept. ☒

1c). Neither / can't tell ☐

2a). The teacher should avoid asking application-based question, without enough context as it will confuse and misdirect the students. ☐

2b). In the lines where the teacher was asking the students to recall examples of indicators, the teacher could have asked the students, ‘How can you determine whether a material is acidic or basic?’ This pushes the children to think beyond simple retention of examples and rather apply their knowledge. ☒

2c). Neither / can't tell ☐

3a). The teacher instructs the students to passively write down definitions. The teacher can instead ask the students to write down their understanding of the concept of indicators. ☒

3b). The teacher should encourage thorough note-taking in the class, as writing down the examples and types of indicators, the students create a concise document for revision of the concepts they learn in class. ☐

3c). Neither / can't tell ☐

Clip B

1. In the clip shown, is there anything the teacher is doing well?

- * Good knowledge of the content
- * Good voice modulation and confidence
- * Taking up the doubts of students

2. In the clip shown, is there anything that the teacher is not doing well?

- * All the students were not involved so it was a bit noisy
- * Some visuals must be involved.
- * Topic/content should be developed with the students' involvement.
- * Not able to seek the interest of all the students.

3. Suggest, if there is any scope of improvement

- * Any model / demonstration / Flash cards / ppt. should be there in support of the content.
- * Students should be involved in content

Clip B

For each question, you will be shown multiple coaching notes written about the same clip. Please read all the options carefully and select **only one** option that, in your judgment, best fits what is actually happening in the specific transcript. There is no requirement that any option is correct. Choose the option that you think best matches the transcript based on your own judgment and expertise. If none of the options can be distinguished as a better fit, select “**Neither / Can’t tell.**”

Please select only one option for each question.

1a). When teacher is introducing activation of deoxyribonucleotides, the teacher could encourage a back and forth discussion with students to bring engagement in classroom. ☒

1b). When teacher is introducing activation of deoxyribonucleotides, the direct, structured way the concept was delivered gives students a clear, uninterrupted line of reasoning to follow. ☐

1c). Neither / can't tell ☐

2a). Throughout the class, teacher could have given relatable analogies and examples to easily relate and visualise the concept of ‘activation of deoxyribonucleotides. ☒

2b). The teacher should spend more time to establish the precise definition, terminologies and mechanisms of deoxyribonucleotide activation concept this will help in further retention. ☐

2c). Neither / can't tell ☐

3a). The teacher shifting from verbal explanation to diagrams without enough verbal framing will confuse the children and their attention might drop. ☒

3b). One thing the teacher should continue doing is, when the teacher drew the diagram of conversion of AMP to ATP, it helped the children's retention of the information by converting a complex process into an equation. ☐

3c). Neither / can't tell ☐

Clip C

1. In the clip shown, is there anything the teacher is doing well?

- * It's a good teacher guided practical session.
- * Teacher is observing and guiding the students to support in learning.

2. In the clip shown, is there anything that the teacher is not doing well?

- * It would be better if the lab had a smart board and the same is running clips for animated/simulated practical before hands on practical/activity.

3. Suggest, if there is any scope of improvement

- * Providing an animated/simulated experience before the actual practical it would have provided more clarity and helped teacher in handling all the students simultaneously.

Clip C

For each question, you will be shown multiple coaching notes written about the same clip. Please read all the options carefully and select **only one** option that, in your judgment, best fits what is actually happening in the specific transcript. There is no requirement that any option is correct. Choose the option that you think best matches the transcript based on your own judgment and expertise. If none of the options can be distinguished as a better fit, select **“Neither / Can’t tell.”**

Please select only one option for each question.

1a). The teacher should continue asking questions such as, “what do you think will happen,” over question that involve recollection as this encourages application of the learned concept rather than passive retention. ☒

1b). The teacher should avoid asking complex questions such as, “what do you think will happen?” and directly state the concept for better retention. ☐

1c). Neither / can't tell ☐

2a). The teacher should avoid asking application-based questions such as “so if this is decreased, what happens to log value?” without enough context as it will confuse and misdirect the students. ☐

2b). The teacher should continue asking application-based questions such as “so if this is decreased, what happens to log value?” which requires drawing inferences to promote higher order thinking. ☒

2c). Neither / can't tell ☐

3a). The teacher should avoid group work and rather work out the problems such as, "if we subtract from this cell the increased value, what do you expect?" in a structured manner to the whole class directly for better retention of the concept. ☐

3b). The teacher should continue promoting group work in students and proposing problems such as, "if we subtract from this cell the increased value, what do you expect?" for them to collectively solve. ☒

3c). Neither / can't tell ☐

Clip A

Correct Notes: 1b, 2b, and 3a

Now that you know which one was generated for this specific lesson, I'd like to ask a couple of questions about that one.

1. How clear was this recommendation to you? 1 = very unclear, 5 = very clear.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒

2. How likely would you be to actually try something like this in your own teaching, or suggest it to a colleague? 1 = very unlikely, 5 = very likely.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒

3. Is there anything about how the notes are worded that felt unnatural, too vague, or too technical? What would you change?

*I feel that its technical based on a proper
Classroom instructional behaviour and teaching
methodology.*

No change is required in measuring tool

Clip B

Correct Notes: 1a, 2a, and 3b

Now that you know which one was generated for this specific lesson, I'd like to ask a couple of questions about that one.

1. How clear was this recommendation to you? 1 = very unclear, 5 = very clear.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒

2. How likely would you be to actually try something like this in your own teaching, or suggest it to a colleague? 1 = very unlikely, 5 = very likely.

1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐

3. Is there anything about how this is worded that felt unnatural, too vague, or too technical? What would you change?

For the response 3 it can involve the students' interest and involvement which could be improved with at least by drawing concept maps, To connect the verbal and visual instructions.

Clip C

Correct Notes: 1a, 2b, and 3b

Now that you know which one was generated for this specific lesson, I'd like to ask a couple of questions about that one.

1. How clear was this recommendation to you? 1 = very unclear, 5 = very clear.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒

2. How likely would you be to actually try something like this in your own teaching, or suggest it to a colleague? 1 = very unlikely, 5 = very likely.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒

3. Is there anything about how this is worded that felt unnatural, too vague, or too technical? What would you change?

It was well framed and technical based on assessment of teaching methodology and no specific changes are required.